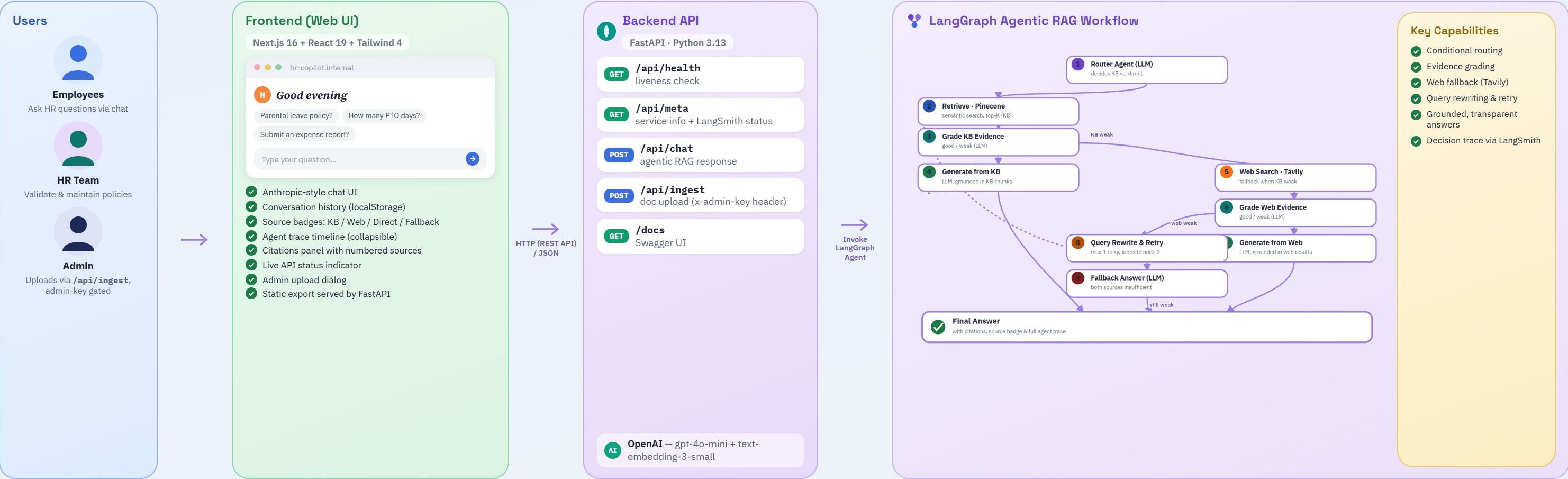


HR Policy RAG Copilot

Agentic Retrieval System — LangGraph + FastAPI + Next.js on DigitalOcean
From Employee Question to Cited Answer



Knowledge Base

DOC = = HR Documents — PDF, DOCX, TXT, MD

Load → Chunk → Embed → Upsert (recursive chunking)

PC **Pinecone Serverless** (AWS us-east-1)

Index: `fde-hr-policy-rag`

Namespace: `company-hr-kb`

Dimensions: 1536 (text-embedding-3-small) · Metric: cosine

External Tools

- TV** **Tavily**: real-time web search, fallback when KB insufficient
- LS** **LangSmith**: execution tracing, LLM-call inspection
- LG** **LangGraph**: orchestrates the agent workflow

Data Layer

- DB** **SQLite**: Chat history / audit · User feedback · Document metadata · Execution trace · System logs
- LS** **localStorage**: conversation history, admin key

Deployment — DigitalOcean + Docker

Single Docker image: multi-stage build

Stage 1: node:20-slim builds Next.js static export

Stage 2: python:3.13-slim serves FastAPI + static

- DO** App Platform · apps-s-1vcpu-0.5gb · \$5/mo
- GH** Auto-deploy from GitHub on push
- SSL** Managed HTTPS (automatic SSL)
- HC** Health check on `/api/health`
- ENV** Env vars + encrypted secrets

Observability

- LangSmith tracing
- DigitalOcean runtime logs
- In-app agent trace panel

Security & Access Control

- 4 encrypted secrets on DO
- x-admin-key on `/api/ingest`
- CORS middleware
- Non-root Docker user
- HTTPS everywhere

Business Outcome

Grounded, cited answers reduce misinformation Fewer repetitive HR queries Transparent reasoning via agent trace

Auto-updates on re-ingest Scalable to hundreds of employees